we want isn't how the model 'thinks' about the concept, and the SAE is working as intended. Alternatively, the SAE training distribution could be too narrow, resulting in the SAE not being incentivised to learn the important latents. (Kissane et al., 2024c) found that SAEs trained on pretraining data generally do not have good latents for the concept of 'refusing' harmful user requests, while SAEs trained on chat formatted data do. Or, the SAE might not have a large enough dictionary size to learn all concepts of interest. Many more hypotheses are plausible. A complicating factor in using SDL to identify the learned mechanisms of neural networks is that the latents identified by depend on the data set used to train them. This is an undesirable property for a decomposition method that was initially hoped to be capable of identifying the fundamental units of computation in neural networks (Kissane et al., 2024c).

2.1.2d Current decomposition methods lack solid theoretical foundations.

Given the practical and conceptual issues with SDL, there is broad agreement that the question of how to correctly decompose networks into atomic units remains a central problem, evidenced by the large amount of effort focused on the direction in recent years. After investing considerable effort in SDL approaches, it is apparent that improved conceptual clarity beyond the idea of superposition (Elhage et al., 2021) is needed to advance neural network decomposition.

One of the most significant open questions is the absence of clarity around the nature of features, despite being the central focus of SDL's identification efforts. Satisfying formal definitions are elusive and conceptual foundations are not yet established. However, even without foundations, progress in mechanistic interpretability is possible — even confused concepts can be pragmatically useful (Henighan, 2024).

Without solid conceptual foundations, it remains unclear whether the superposition hypothesis, which underpins the SDL paradigm, is fundamentally valid or merely pragmatically useful (Henighan, 2024; Templeton et al., 2024). If it is the latter, there may be better methods than SDL for carving neural networks at their joints. Such methods may take into account feature geometry or take a more dynamic, developmental view of how mechanistic structure emerges in the training process (Hoogland et al., 2024; Wang et al., 2024b). Moreover, although there is much emphasis on how models represent features in superposition, there is comparatively less work examining how they might perform computation on them natively in superposition. Further conceptual work into this problem may suggest new methodologies for decomposing networks, or provide bounds on the number of features we should expect models to be capable of learning (Hänni et al., 2024; Adler & Shavit, 2024; Bushnaq & Mendel, 2024).

Mechanistic interpretability should ideally be built on more formal foundations. Some work attempts to ground mechanistic interpretability in the formalisms of causality (Geiger et al., 2024a). However, this approach has not yet yielded canonical causal mediators (Mueller et al., 2024) that may form a basis for decomposition methods. How should we go about finding them? Given that our field's objective is to understand the learned structures that underlie networks' generalization behaviors, exploring theories about why neural networks generalize appears to be a promising avenue. But theories that attempt to characterize why neural networks generalize, such as the spline theory of neural networks (Balestriero & richard baraniuk, 2018), theories of neural networks' simplicity bias (Valle-Perez et al., 2019), deep learning theory involving the neural tangent kernel (Jacot et al., 2018; Roberts et al., 2022), or singular learning theory (Watanabe, 2009; Wei et al., 2023) have either not yet yielded mathematical objects that can be easily used for interpretability, or simply have not been successfully linked to approaches for interpreting neural networks. Establishing these connections would make significant progress toward carving neural networks at their joints to facilitate mechanistic interpretability. And if we can carve trained networks at their joints, it may suggest ways to train networks such that they come 'pre-carved'. Thus, better theoretical foundations may also be important for developing models that are intrinsically decomposable by design, which we discuss next.

2.1.2e Intrinsic interpretability: Building more easily decomposable models

The current strategy of training a model solely for performance and then interpreting it post hoc may not be optimal if our goal is a model that is both interpretable and performant. To this end, it may be beneficial to prioritize interpretability during model training, for which there are several plausible approaches.

Instead of post-hoc decomposing trained network activations into discrete codes (Paragraph 2.1.2c), network activations could be forced to use a discrete code from the outset, as in Tamkin et al. (2025). MLPs could also be trained with sparser activation functions, such as TopK (Makhzani & Frey, 2013; Bills et al., 2023) or SoLU (Elhage et al., 2022). Similar approaches could be potentially used to restrict attention superposition (Jermyn et al., 2023) by limiting the number of heads attending to any query-key pair. Several approaches, such as 'mixture of experts' (Shazeer et al., 2017; Fedus et al., 2022; He, 2024), use sparsely activating components — with a large enough number of experts, and with sufficient activation sparsity, experts may become individually interpretable (Warstadt et al., 2019). Many attempts to incentivize interpretable activations directly so far have not been competitively performant, and have also allowed 'superposition to sneak through', mitigating benefits.

We can also target weight sparsity directly during training, including approaches such as L0 regularization (Louizos et al., 2018) or pruning (Mozer & Smolensky, 1988; Frankle & Carbin, 2019; Mocanu et al., 2018). Han et al. (2015) achieve sparse weights by using magnitude pruning followed by finetuning. This highlights a general strategy of finetuning with interpretability in mind (also used by Tamkin et al. (2025)), avoiding the potentially excessive cost of training from scratch. Another related strategy is targeting modularity (Kirsch et al., 2018; Andreas et al., 2016). For example, brain-inspired modular training (Liu et al., 2023) trains for modularity by embedding neurons in a geometric space and encouraging geometrically local connections.

Existing research implements other approaches that can simplify linearity-reliant circuit analysis (Elhage et al., 2021). For example, there is work removing the layer norm operations (Heimersheim, 2024), using input-switched affine transformations for recurrence (Foerster et al., 2017). Other studies leverage architectures that are mathematically analyzable in other ways than linearity, such bilinear activations in MLPs (Sharkey, 2023; Pearce et al., 2024).

After decomposing a network into components, whether through post hoc decomposition or by using intrinsically decomposable models, our task remains unfinished. We must provide an interpretation of the functional role of each component (Step 2 – Section 2.1.3) and validate that interpretation (Step 3 – Section 2.1.4). In the next section, we will discuss common methods of interpretation and their shortcomings.

2.1.3 Reverse engineering step 2: Describing the functional role of components

After decomposing networks into parts, the next step of reverse engineering is to "describe" the functional role of these components. This step is best thought of as generating hypothesized 'interpretations' or 'explanations'. These explanations form candidate descriptions of the functional role of a given component, and should not be taken to be definitive conclusions before they are thoroughly validated (see Section 2.1.4).

Descriptions of the functional role of neural network components can either indicate (1) the cause of a component's activation or (2) what occurs after that component has been activated, or – preferably – both (Figure **5**). In this section, we will discuss the existing set of tools available to mechanistic interpretability researchers to describe the functional role of network components.

2.1.3a Explanations for what causes components to activate

Explanations of the causes of component activation can use three broad categories of methods, each with several problems: (1) Highly activating data set examples; (2) Attribution methods; and (3) Feature synthesis.

Highly activating data set examples. The simplest method is to use highly activating data set examples (sometimes called 'exemplar representations' (Hernandez et al., 2022)). These are inputs on which a particular component is strongly activated. For a given component, analyzing apparent commonalities in the inputs suggests hypotheses for what causes that component to activate. This step may be carried out by humans or AI systems (Section 2.4).

Despite being widely used, this approach has several substantial issues. The first issue is that the method relies on human prior beliefs, which may lead interpreters to project their human understanding onto models that may, in fact, be using unfamiliar concepts. This bias could lead us to identify concepts in the model

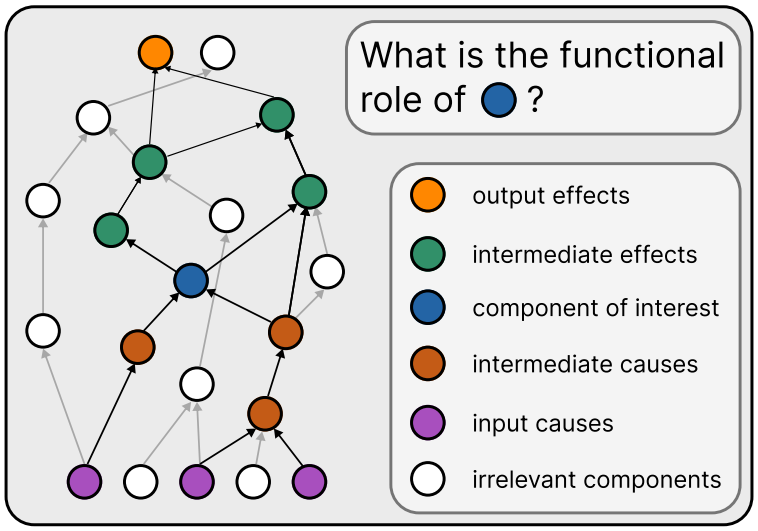

Figure 5: To study the functional role of the blue component numerous approaches are possible. We could study its causes: the purple input components or the red intermediate components via e.g. feature synthesis (Olah et al., 2017a; 2020b), maximum activating examples (Olah et al., 2017a; Bricken, 2023), or attributions (Sundararajan et al., 2017). Or we could study its effects: the orange output components or the green intermediate components via e.g. the logit lens (Nostalgebraist, 2020), activation steering (Turner et al., 2024), or attributions (Marks et al., 2024).

that do not truly explain the mode's functioning (Freiesleben & König, 2023; Donnelly & Roegiest, 2019; Gale et al., 2020).

Another issue with this approach is the potential for 'interpretability illusions'. Bolukbasi et al. (2021) show that bias in data sets can create misleading explanations even when top activating examples are selected from real data sets, as opposed to synthetically created data. Depending on the data set from which the examples were drawn, human annotators identified dramatically different meanings for given directions in the activation space of BERT (Devlin, 2018).

A third issue is that this approach often yields plausible explanations for arbitrarily chosen directions in the activation space (Szegedy et al., 2014). This means that plausible explanations based on highly activating data set examples cannot be solely relied upon to identify the basic units of computations in neural networks — other methods are needed to accurately identify them. Furthermore, it is possible to develop adversarial models that deliberately yield misleading feature visualizations (Geirhos et al., 2025).

Many of the issues with highly activating data set examples stem from the fact that they merely provide correlational explanations for the activation of a network component, rather than causal explanations. To identify causal explanations, other methods, such as attribution methods, are necessary.

Attribution methods are necessary for causal explanations but are often difficult to interpret. Attribution methods (Simonyan et al., 2014a; Nguyen et al., 2016b; Selvaraju et al., 2019; Sundararajan et al., 2017; Fong & Vedaldi, 2017; Lundberg & Lee, 2017; Ribeiro et al., 2016) are intended to measure the causal importance of upstream variables (such as inputs) on downstream variables (such as a network component). They are often gradient-based (Mozer & Smolensky, 1988; Simonyan et al., 2014a; Nguyen et al., 2016b; Selvaraju et al., 2019; Sundararajan et al., 2017; Wang et al., 2024d) or sampling-, perturbation-, or ablation-based (Fong & Vedaldi, 2017; Vig et al., 2020; Geiger et al., 2020; Ghorbani & Zou, 2020; Meng et al., 2022b; Chan et al., 2022a; Nanda, 2023a). However, on a theoretical level, many gradient-based methods identify only a first-order approximation of the ideal attribution, which is sometimes a poor approximation (Watson, 2022). Adebayo et al. (2018) revealed more practical implications, demonstrating that some gradient-based methods identify attributions that are independent both of the model and of the data generating process. Furthermore, an adversary can train a model or perturb an input to reveal any